ECON2103 Microeconomics

Chapter 9 The Analysis of Competitive Markets

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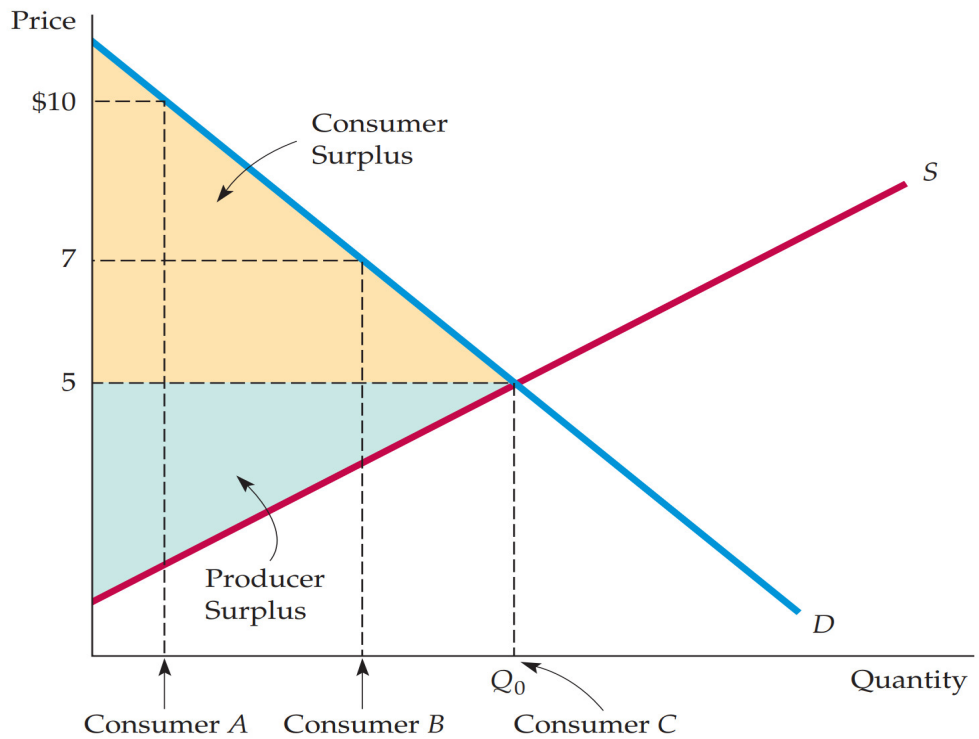


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Outline

1. Evaluating the Gains and Losses
2. The Efficiency of a Competitive Market
3. Minimum Prices
4. Price Supports and Production Quotas
5. Import Quotas and Tariffs
6. The Impact of a Tax or Subsidy
7. Summary and Exercises

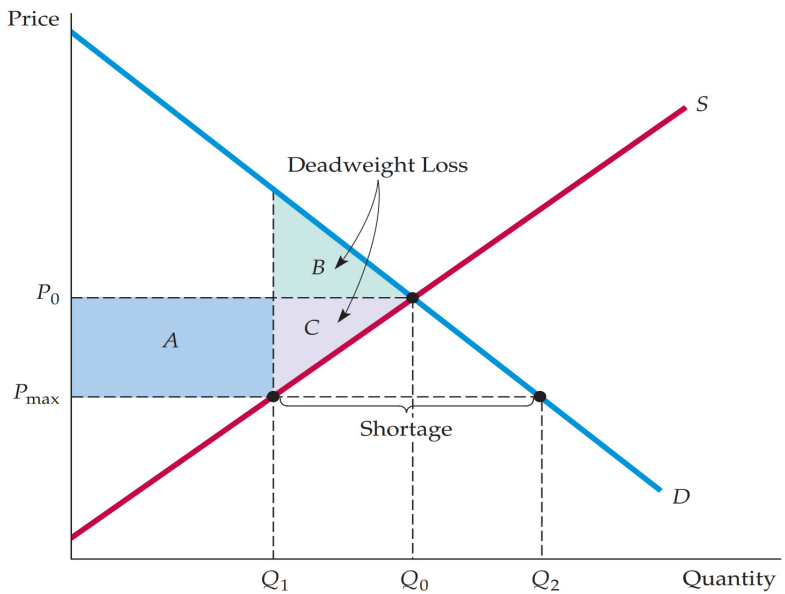
Review of Consumer and Producer Surplus



Consumer and producer surplus measure the welfare benefit of a competitive market.

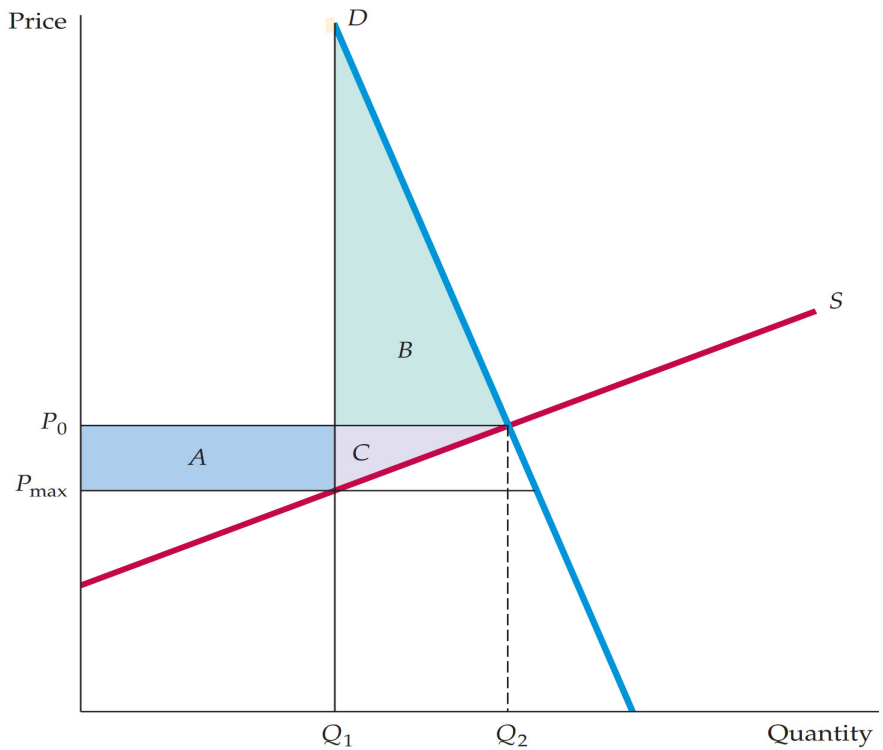
Application of Consumer and Producer Surplus

welfare effects - Gains and losses to consumers and producers.
deadweight loss - Net loss of total (consumer plus producer) surplus.



- The price of a good has been regulated to be no higher than P_{max}.
- The gain to consumers is the difference between rectangle A and triangle B.
- The loss to producers is the sum of rectangle A and triangle C.
- Triangles B and C together measure the deadweight loss from price controls.

Effect of Price Controls When Demand is Inelastic

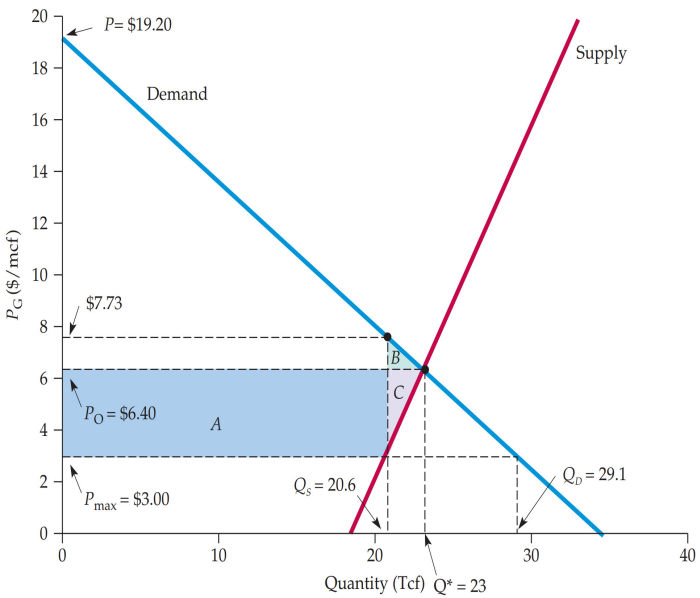


If demand is sufficiently inelastic, triangle B can be larger than rectangle A. In this case, consumers suffer a net loss from price controls.

Example: Effects of Natural Gas Price Controls

$$\text{Supply: } Q_S = 15.90 + 0.72P_G + 0.05P_O$$

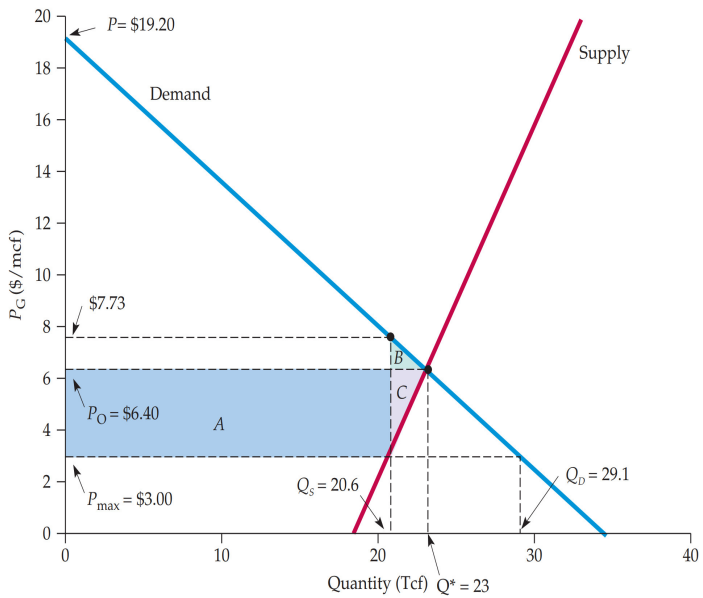
$$\text{Demand: } Q_D = 0.02 - 1.8P_G + 0.69P_O$$



- The market-clearing price of natural gas was \$6.40 per mcf, and the (hypothetical) maximum allowable price is \$3.00.
- A shortage of $29.1 - 20.6 = 8.5$ Tcf resuults.

Example: Effects of Natural Gas Price Controls (Con't)

- $A = 20.6(3.40) = \$70.04$ billion.
- $B = (1/2)(2.4)(1.33) = \$1.60$ billion.
- $C = (1.2)(2.4)(3.40) = \$4.08$ billion.
- The annual change in consumer surplus would be $A - B = 70.04 - 1.60 = \$68.44$ billion.
- The change in produce surplus would be $-A - C = -70.04 - 4.08 = -\$74.12$ billion.
- The annual deadweight loss would be $-B - C = -1.60 - 4.08 = -\$5.68$ billion.

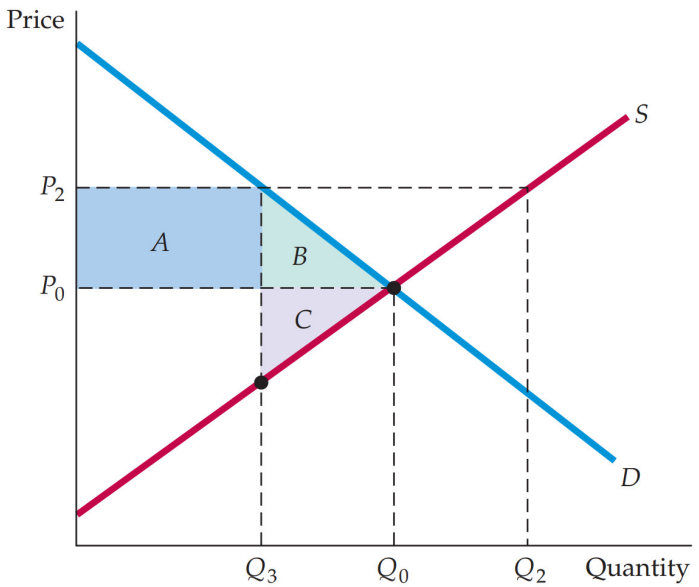


Economic Efficiency

- **economic efficiency** - Maximization of aggregate consumer and producer surplus.
- **market failure** - Situation in which an unregulated competitive market is inefficient because prices fail to provide proper signals to consumers and producers. There are two important instances in which market failure can occur:
  1. Externalities
     - **externality** - Action taken by either a producer or a consumer which affects other producers or consumers but is not accounted for by the market price.
  2. Lack of information
     - Market failure can also occur when consumers lack information about the quality or nature of a product and so cannot make utility-maximizing purchasing decisions. Government intervention may then be desirable.

Welfare Loss When Price is Held Above Market-Clearing Level

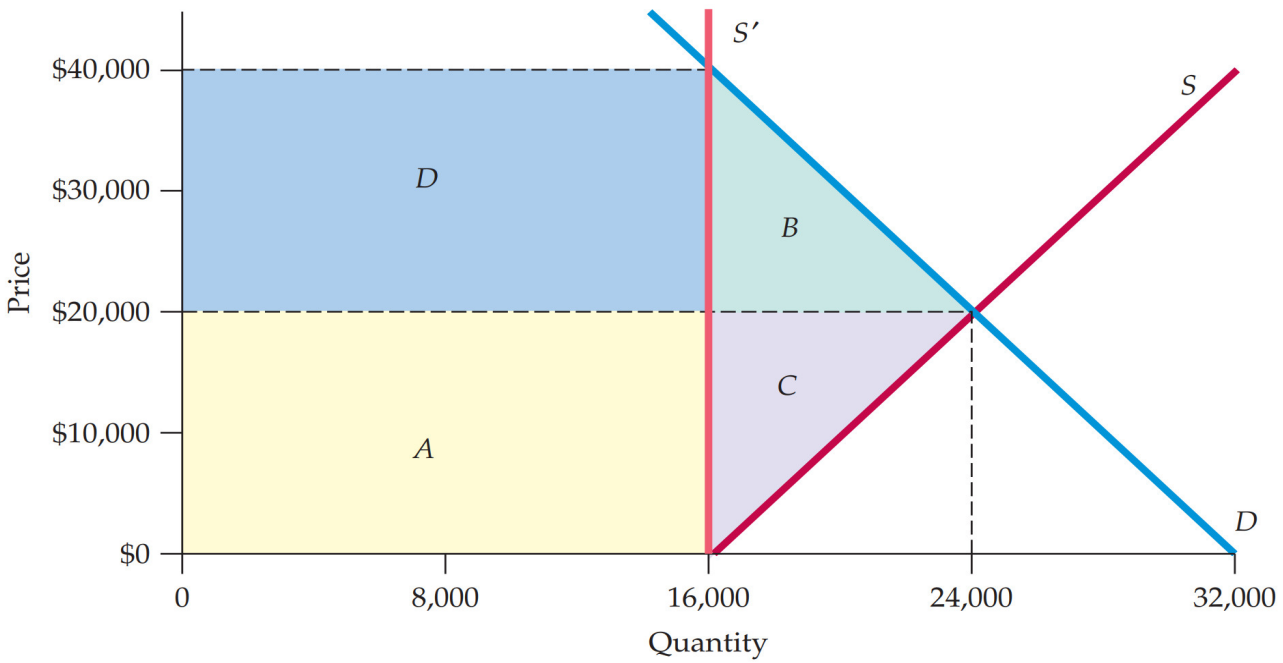
- When price is regulated to be no lower than P_2, only Q_3 will be demanded.
- If Q_3 is produced, the deadweight loss is given by triangles B and C.



Example: The Market for Human Kidneys

Supply: $Q^S = 16,000 + 0.4P$

Demand: $Q^D = 32,000 - 0.4P$

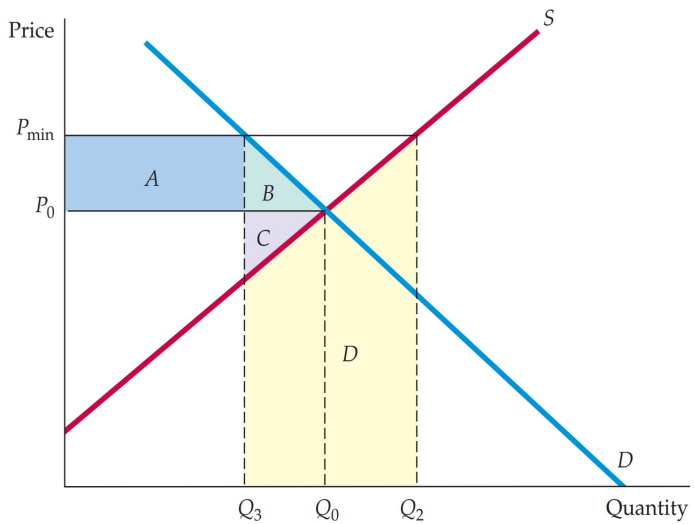


Example: The Market for Human Kidneys (Con't)

- The market-clearing price is \$20,000; at this price, about 24,000 kidneys per year would be supplied.
- The law effectively makes the price zero. About 16,000 kidneys per year are still donated; this constrained supply is shown as S'.
- The loss to suppliers is given by rectangle A and triangle C.
- If consumers received kidneys at no cost, their gain would be given by rectangle A less triangle B.

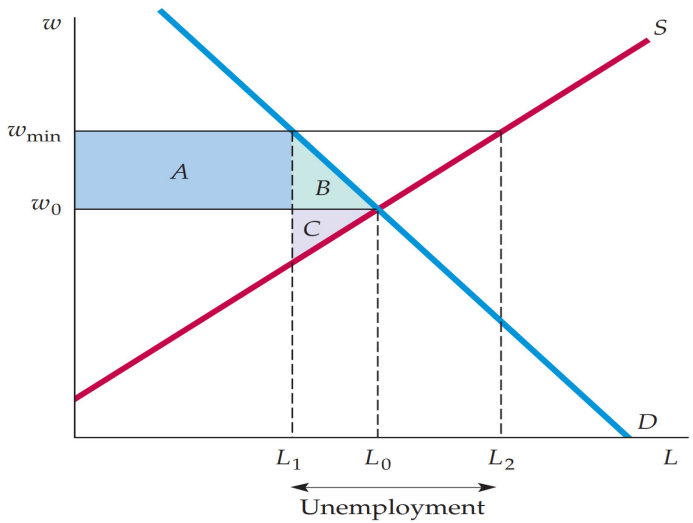
Price Minimum

- Price is regulated to be no lower than P_{min}.
- Producer would like to supply Q_2 but consumers will buy only Q_3.
- The total change in consumer surplus is: $\Delta CS = -A - B$.
- The total change in producer surplus is: $\Delta PS = A - C - D$.



The Minimum Wage

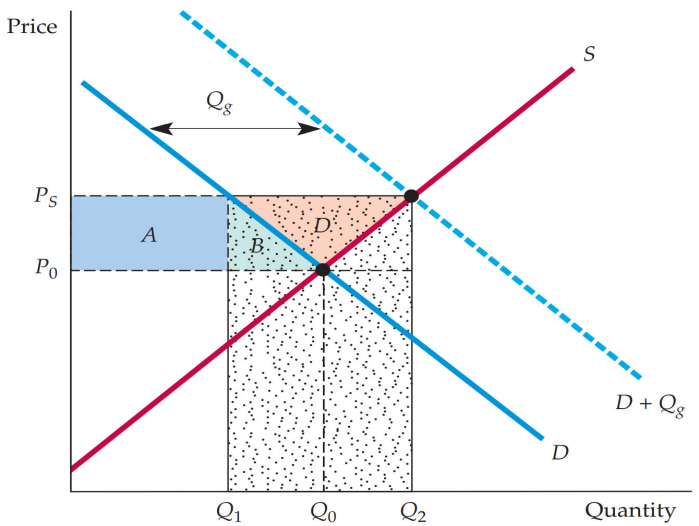
- Although the market-clearing wage is w_0, firms are not allowed to pay less than w_{min}.
- This results in unemployment of an amount $L_2 - L_1$ and a deadweight loss given by triangles B and C.



Price Supports

price support - Price set by government above free-market level and maintained by governmental purchases of excess supply.

- To maintain a price P_S above the market-clearing price P_0, the government buys a quantity Q_g.
- The gain to producers is $A+B+D$. The loss to consumers is $A+B$.
- The cost to the government is the speckled rectangle, the area of which is $P_S(Q_2-Q_1)$.



Price Support (Con't)

- **Consumers**: Some consumers pay a higher price, while others no longer buy the good

$$\Delta CS = -A - B$$

- **Producers**: Producers are now selling a larger quantity Q_2 instead of Q_0, and at a higher price P_S

$$\Delta PS = A + B + D$$

- **The government**: The cost to the government is

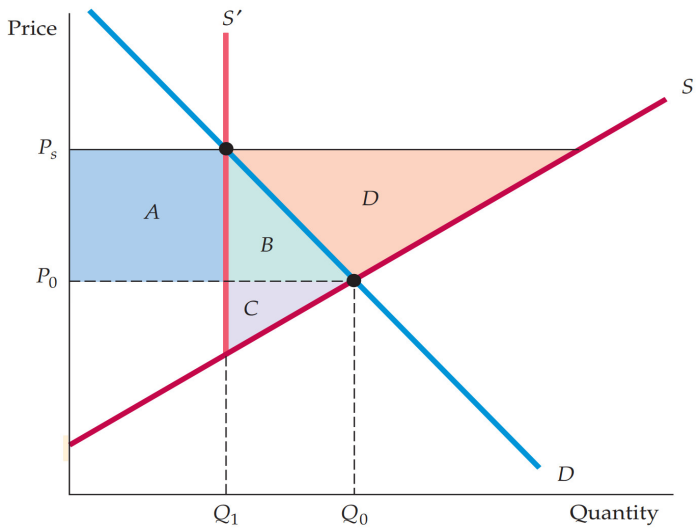
$$(Q_2 - Q_1)P_S$$

The total change in welfare is

$$\Delta CS + \Delta PS - \text{Cost to Govt.} = D - (Q_2 - Q_1)P_S$$

Production Quotas

- To maintain a price P_S above the market-clearing price P_0, the government can restrict supply to Q_1, either by imposing production quotas or by giving producers a financial incentive to reduce output.
- The government restricts the quantity supplied to Q_1. The supply curve becomes S'.
- Consumer surplus is reduced by rectangle A plus triangle B.
- Producers gain rectangle A but lose triangle C.
- There is a deadweight loss, given triangles B and C.



Incentive Programs

Output can be reduced by financial incentives rather than by outright quotas.

- As with direct production quotas, the change in consumer surplus is

$$\Delta CS = -A - B$$

- Producers receive a higher price, produce less, and receive an incentive to reduce production. Thus, the change in producer surplus is

$$\Delta PS = A - C + \text{Payments for not producing}$$

- The cost to the government is at least $B + C + D$. Thus

$$\Delta PS = A - C + B + C + D = A + B + D$$

- The total change in welfare is

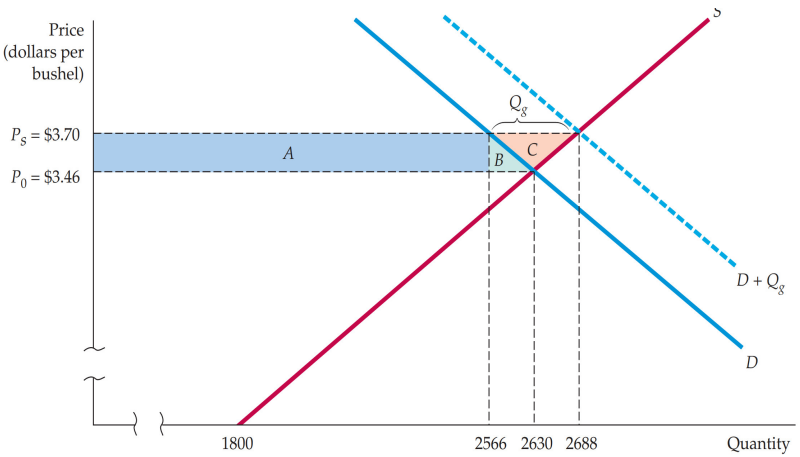
$$\Delta \text{Welfare} = -A - B + A + B + D - B - C - D = -B - C$$

Example: Supporting the Price of Wheat

$$\text{1981 Supply: } Q_S = 1800 + 240P$$

$$\text{1981 Demand: } Q_D = 3550 - 266P$$

- To increase the price to \$3.70, the government must buy the quantity of wheat Q_g.
- 1981 Total demand: $Q_{DT} = 3550 - 266P + Q_g$. Then $Q_g = 506P - 1750 = 122$ million bushels
- Loss to consumers = $-A - B = \$624$ million.
- Cost to the government = $3.70(122) = \$451.4$ million.
- Total cost of the program = $\$624 + \$451.4 = \$1075$ million.
- Gain to producers = $A + B + C = \$638$ million.

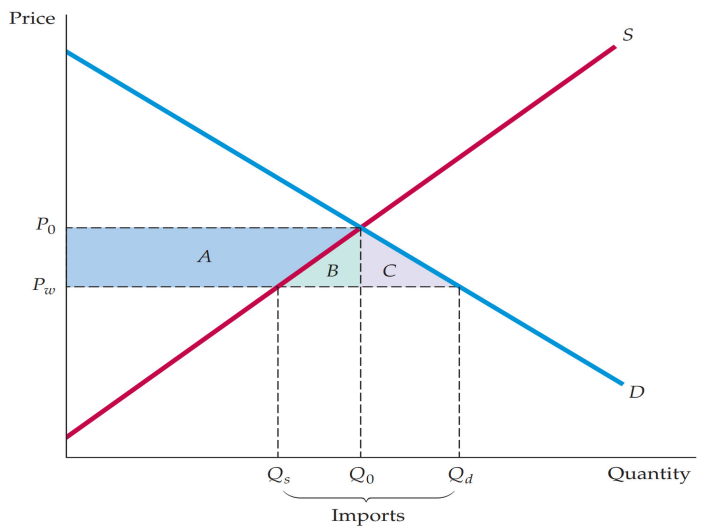


Import Quotas and Tariffs

import quota - Limit on the quantity of a good that can be imported.

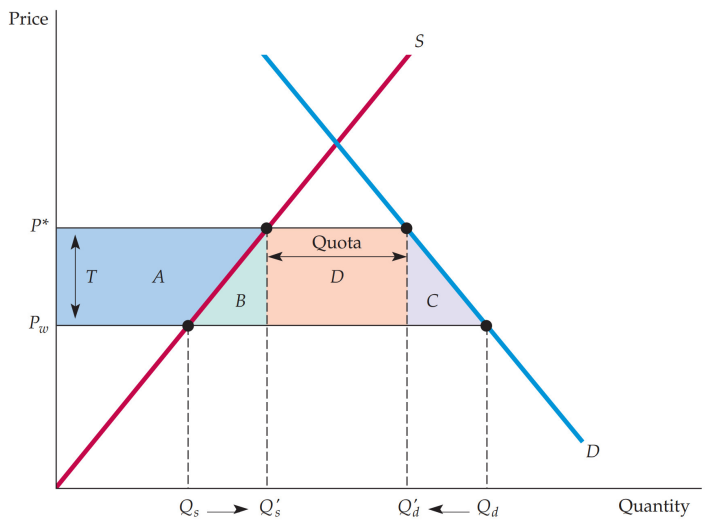
tariff - Tax on an imported good.

- In a free market, the domestic price equals the world price P_w.
- A total Q_d is consumed, of which Q_s is supplied domestically and the rest imported.
- When imports are eliminated, the price is increased to P_0.
- The gain to producers is A.
- The loss to consumers is $A+B+C$, so the deadweight loss is $B+C$.



Import Quotas and Tariffs (Con't)

- When imports are reduced, the domestic price is increased from P_w to P^*. This can be achieved by a quota, or by a tariff $T = P^* - P_w$.
- Trapezoid A is the gain to domestic producers.
- The loss to consumers is $A + B + C + D$.
- If a tariff is used, the government gains D, the revenue from the tariff. The net domestic loss is $B + C$.
- If a quota is used, rectangle D becomes part of the profits of foreign producers, and the net domestic loss is $B + C + D$.

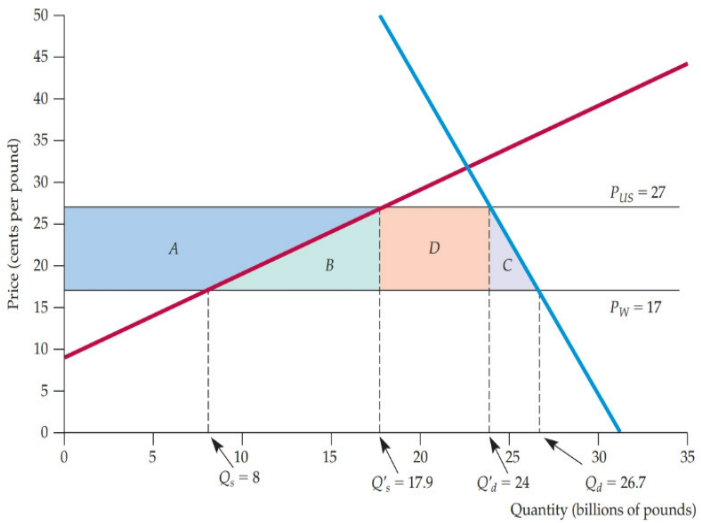


Example: The Sugar Quota

$$\text{U.S. supply: } Q_S = -8.95 + 0.99P$$

$$\text{U.S. demand: } Q_D = 31.20 - 0.27P$$

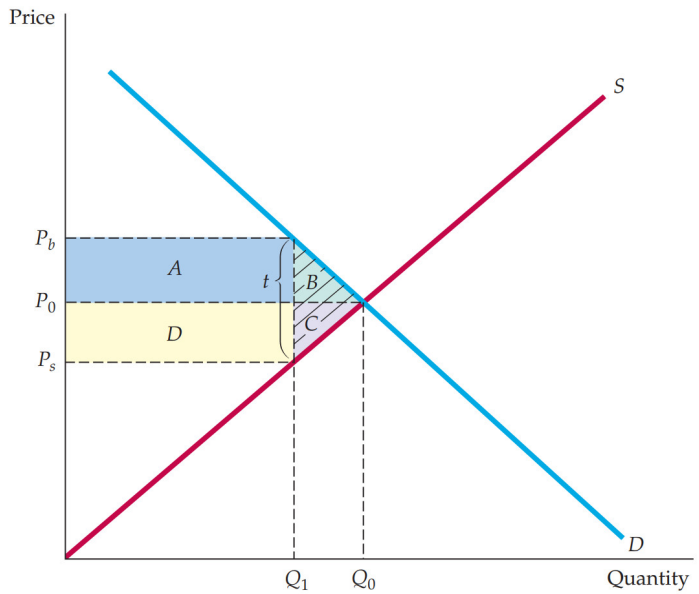
- At the world price of 17 cents per pound, about 26.7 billion pounds of sugar would have been consumed in 2016, of which all but 8 billion pounds would have been imported.
- Restricting imports to 6.1 billion pounds caused the U.S. price to go up by 10 cents.
- The cost to consumers, $A + B + C + D$, was about \$2.5 billion.
- The gain to domestic producers was A, about \$1.3 billion.
- Rectangle D, \$610 million, was a gain to those foreign producers who obtained quota allotments.
- Triangles B and C represent the deadweight loss of about \$631 million.



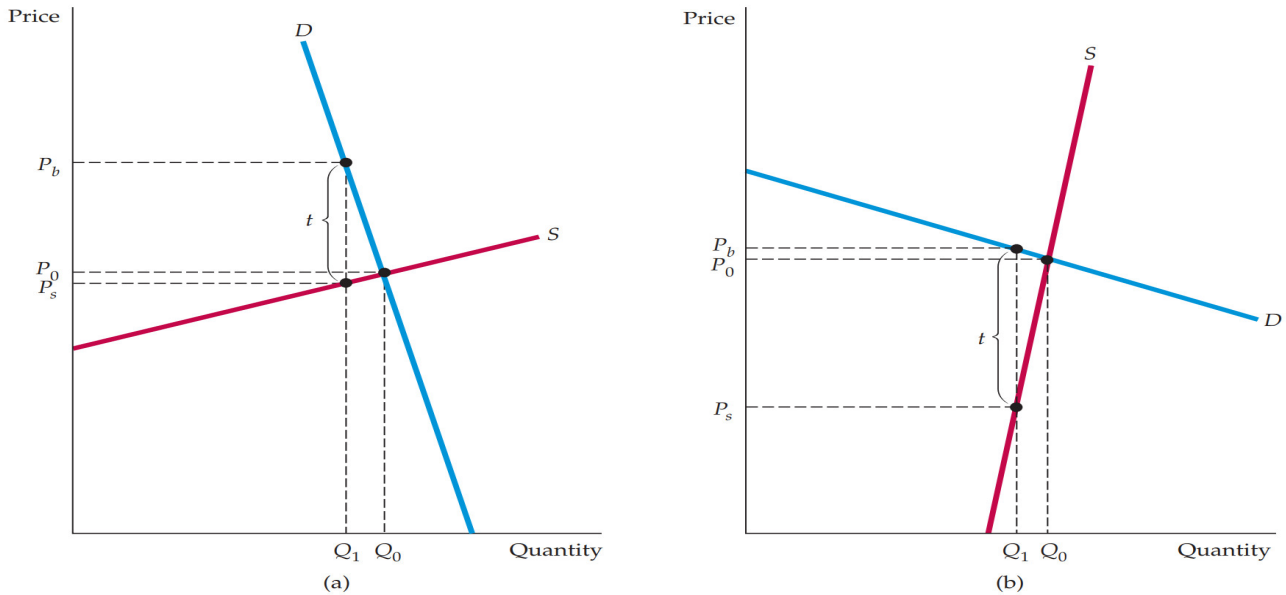
The Effets of a Specific Tax

specific tax - Tax of a certian amount of money per unit sold.

- P_b is the price (including the tax) paid by buyers. P_s is the price that sellers receive, less the tax.
- Buyers lose $A + B$.
- Sellers lose $D + C$.
- The government earns $A + D$ in revenue.
- The deadweight loss is $B + C$.



Impact of a Tax Depends on Elasticities of Supply and Demand

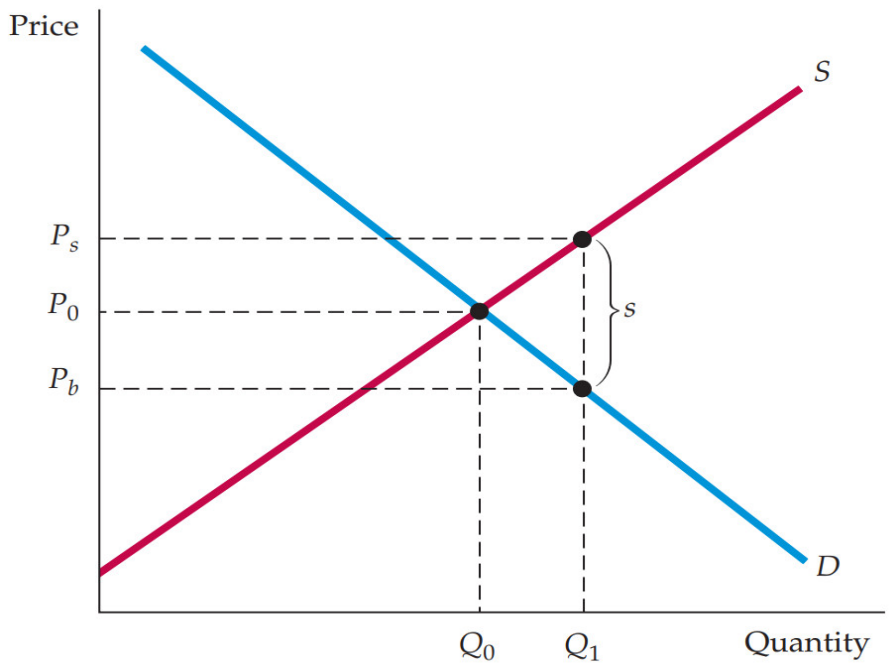


- **(a)** If demand is very inelastic relative to supply, the burden of the tax falls mostly on buyers.
- **(b)** If demand is very elastic relative to supply, it falls mostly on sellers.
- By using the following "pass-through" formula, we can calculate the percentage of the tax that is "passed through" to consumers: Pass through fraction $= E_s/(E_s - E_d)$

The Effects of a Subsidy

subsidy - Payment reducing the buyer's price below the seller's price; i.e., a negative tax.

Like a tax, the benefit of a subsidy is split between buyers and sellers, depending on the relative elasticities of supply and demand.



Example: A Tax on Gasoline

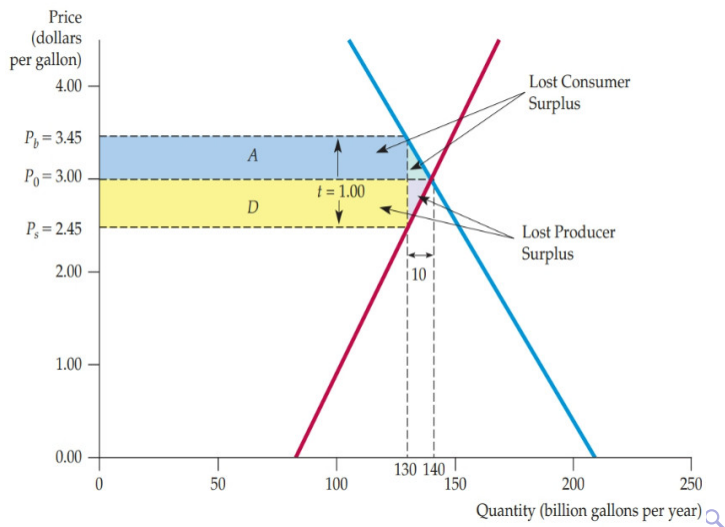
$$\text{Demand: } Q^D = 209 - 23P_b$$

$$\text{Supply: } Q^S = 83 + 19P_s$$

$$\text{Supply must equal demand: } Q^D = Q^S$$

$$\text{Government must receive \$1.00/gallon: } P_b - P_s = 1.00$$

- $P_s = 2.45, P_b = 3.45, Q = 130$
- Annual revenue from the tax: $tQ = (1.00)(130) = \$130$ billion.
- Deadweight loss: (1/2)(1.00/gallon)(10 billion)=\$5 billion per year.



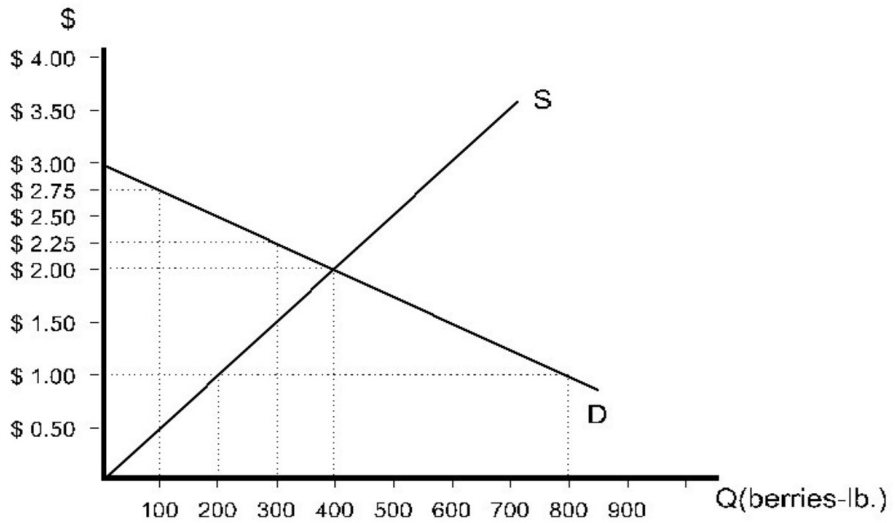
Summary

- Evaluating gains and losses
  - welfare effects and deadweight loss
- Efficiency of a competitive market
  - welfare loss
- Minimum prices
- Price supports and production quotas
- Import quotas and tariffs
- Impact of a tax and subsidy

Exercises

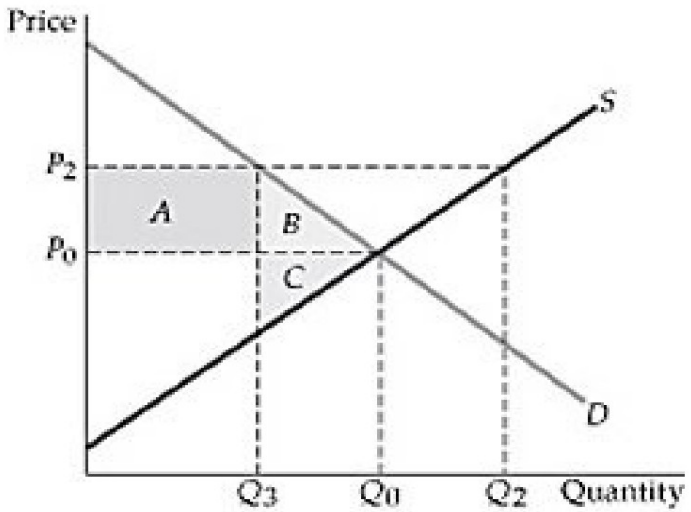
1. An effective price ceiling causes a loss of (a)
   a. producer surplus for certain and possibly consumer surplus as well.
   b. consumer surplus only.
   c. producer surplus only.
   d. consumer surplus for certain and possibly producer surplus as well.
   e. neither producer nor consumer surplus.

Exercises (Con't)



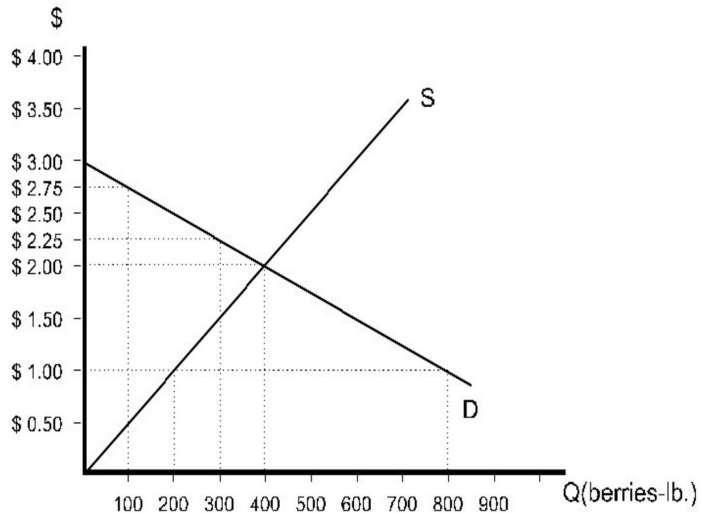
2. If the market is in equilibrium, the consumer surplus earned by the buyer of the 100th unit is (b)
   a. $0.50.
   b. $0.75.
   c. $1.50.
   d. $2.00.
   e. $2.75.

Exercises (Con't)



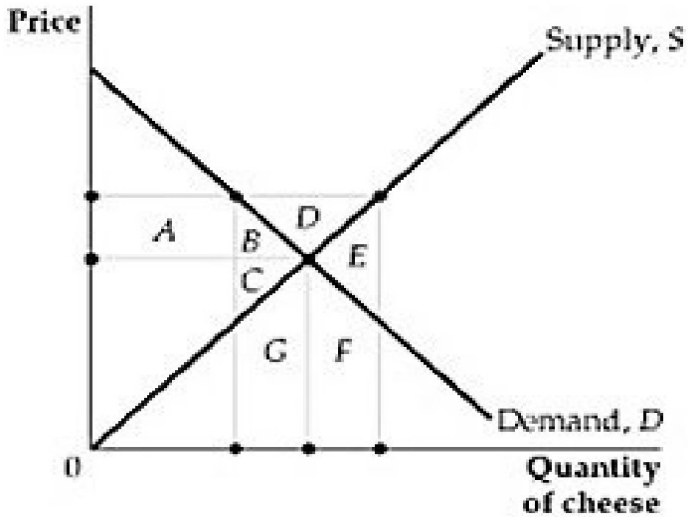
3. After the minimum price P_2 is imposed, (c)
   a. some consumers are better off and all producers are worse off.
   b. some consumers and some producers are better off.
   c. no consumers are better off but some producers are better off.
   d. no consumers are better off and all producers are better off.

Exercises (Con't)



4. If the government establishes a price floor of \$2.50 and farmers grow only the amount of berries that will be sold, producer surplus will: (c)
   a. fall by \$50.
   b. fall by \$100.
   c. remain the same.
   d. rise by \$50.
   e. rise by \$100.

Exercises (Con't)

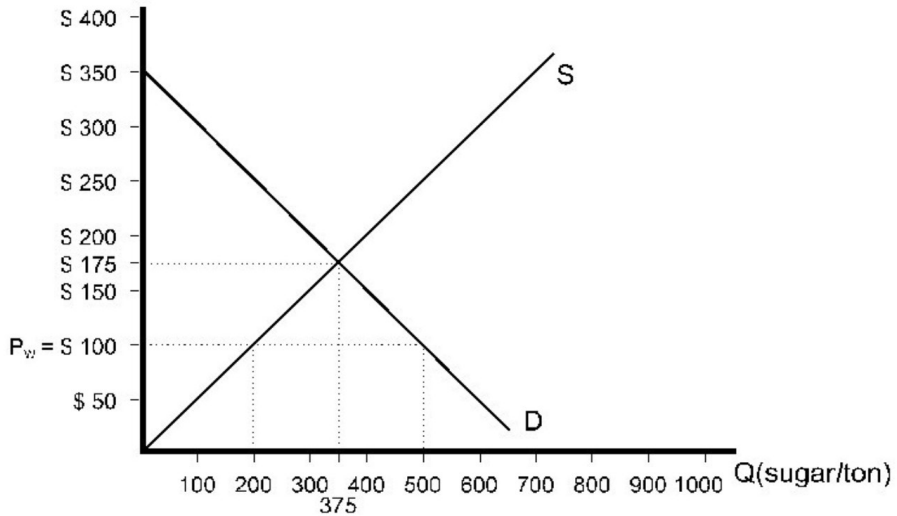


5. Suppose the government raises the price of cheese above the market equilibrium level (P_0) by imposing a high minimum price and purchasing all of the excess supply from the market, and these quantities are destroyed. Based on the areas in the figure below, what is the change in producer surplus after this policy is adopted? (d)
   a. Producers lose area C but gain area A.
   b. Producers lose area C buy gain area A.
   c. Producers gain A.
   d. Producers gain area A+B+D.

Exercises (Con't)

6. What is the difference between a price support and a price floor? (d)
   a. A price support is below equilibrium; a price floor is above it.
   b. A price support is above equilibrium; a price floor is below it.
   c. Government buys the excess supply to maintain a price floor, but not a price support.
   d. Government buys the excess supply to maintain a price support, but not for a price floor.
   e. There is no difference between the two.

Exercises (Con't)



7. A $50 tariff would result in domestic consumption of: (c)
   a. 600, domestic production of 100, and imports of 500.
   b. 500, domestic production of 200, and imports of 300.
   c. 400, domestic production of 300, and imports of 100.
   d. 300, domestic production of 400, and exports of 100.
   e. 200, domestic production of 500, and exports of 300.

Exercises (Con't)

8. Compared to a tariff, an import quota, which restricts imports to the same amount as the tariff, will leave the country as a whole: (a)
   a. worse off than a comparable tariff.
   b. not as bad off as a comparable tariff.
   c. about the same as a comparable tariff.
   d. Any of the above can be true.

Exercises (Con't)

9. The benefit of a subsidy accrues mostly to consumers: (c)
   a. in every instance.
   b. if E_d/D_s is large.
   c. if E_d/E_s is small.
   d. if E_d and E_s are equal.
   e. in no instance.

Exercises (Con't)

10. When the government imposes a specific tax per unit on a product, changes in consumer surplus are _____ and changes in producer surplus are _____. (c)
    a. negative; positive
    b. positive; positive
    c. negative; negative
    d. positive; negative